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## Detection of tooth decay and its severity based on processing of panoramic radiographic images

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# Detection of tooth decay and its severity based on processing of panoramic radiographic images

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## Abstract

X-ray dental imaging helps dentists and radiologists diagnose dental disease and provide treatment plans for patients. In many cases, it is difficult to diagnose dental disease based on visual inspection alone. Therefore, the development of a support system for caries detection has been an issue in recent years. The most challenging type of image for this process is the Panoramic image. Because it includes other parts of the patient's mouth in addition to the teeth, the structures also have no clear boundaries. Here, we propose an automated detection system to detect tooth decay in full view images. The proposed model uses convolutional neural network to extract related features from X-ray. In this paper images are obtained from the Diagnostic Imaging Center of the Southwest State University of Bahia (UESB). 1,000 images of this dataset was manually tagged and 750 images were used as training data and the rest as test dataset. Our model achieved an accuracy score of 94% in the experimental set. Scores show acceptable diagnostic performance as well as speed up caries diagnosis, reducing the challenge of using a full-length photograph of real patients. Among the images with caries lesions in the experimental set, our model scored 64% and 69% for mild and severe cases, confirming the fact that severe caries spots are easier to diagnose and mild caries detection is more efficient. A stronger system is needed. This is a step towards developing an efficient, fully automated decision support system to assist experts in the field.

## CCS Concepts

• Software and its engineering;

## Keywords

Dental caries diagnosis, complete dental radiograph, neural network

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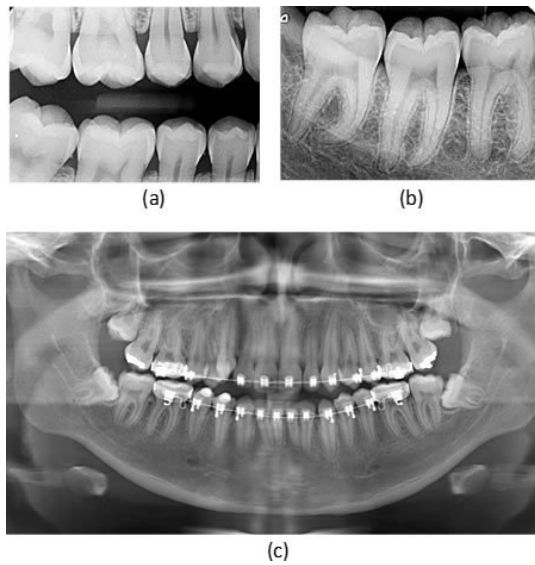
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## 1 Introduction (Teeth and dental care)

Tooth decay is one of the chronic diseases that most people deal with during their lifetime [1]. According to the National Health and Nutrition Examination Survey, tooth decay affects approximately 90% of adults in the United States of America [2], [3]. The main method for diagnosing dental caries is clinical examination, and radiographic examination is a complementary diagnostic tool [4]. Dental anatomy is a field of anatomy that is dedicated to the study of the structure of human teeth. Growth, appearance and classification of teeth are within its authority. Most teeth have distinctive features. Dental plaque is a dental disease caused by the complex interactions of starch and sugar supplied through food. This plaque produces acids that lead to the decay of minerals in the enamel area. If a simple case of caries on the surface of the tooth protein left untreated, leads to the formation of deep cavities with bacterial infection in the pulp areas. These cavities or bacterial infection mostly lead to infection in the gum tissues.

### 1.1 Types of dental radiographic images

A standard tool used by radiologists in the diagnosis of dental diseases such as caries is X-ray radiography [5]. X-ray radiographs in dentistry have types of intraoral (such as bitewing and periapical) and extra oral (such as Orthopantomogram (OPG) photos), each of which shows a different view of the tooth anatomy. Basically, bitewing radiography is the most widely used method for the diagnosis of caries and has the highest diagnostic accuracy, while the Panoramic photograph has the lowest diagnostic accuracy [6, 7]. However, intraoral radiographic images have several disadvantages, including discomfort for the patient. Also, usually due to the insufficient expertise of the operators and the renewal of photography, these photos lead to an increase in the radiation dose of the patient [8]. A Panoramic image, also called an OPG images, provides a wider picture of the entire mouth using a very small dose of X-rays. Also, its simplicity, greater comfort for the patient and high speed of taking pictures are its other advantages. This type of photo is more suitable for special patients such as children, the disabled and the elderly ones compared to intraoral systems [9]. But at the same time, panoramic photo analysis comes with challenges. For example, the structures in full-view images do not have clear boundaries and their visual quality is very low compared to other types of dental radiographs. In addition, panoramic images include other parts of the mouth such as the jawbone, which makes image analysis difficult. Therefore, in order to facilitate the diagnosis of dental disease from panoramic photos, performing image pre-processing steps and single tooth isolation is a mandatory procedure [6]. Fig. 1 shows examples of radiographic images of teeth.



**Figure 1: a) bitewing photo b) periapical photo c) panoramic photo [4].**

## 1.2 Technology and the challenge of detecting dental caries

In the past two decades, computer-aided diagnosis has developed significantly. In dentistry, the processing of radiographic images has also become an important topic of automation, as radiographic image interpretation is an essential part of diagnosis, dental health monitoring, and treatment planning.

During the last decade, a number of studies have addressed this problem. For tooth detection, Lin et. al [20] and Hosntalab et al [21] proposed pixel-level segmentation methods based on traditional computer vision techniques, such as histogram-based thresholding methods, and surface ensemble methods. To solve the mentioned challenges and problems, a limited number of single tooth extraction researches from panoramic images using different methods, such as deep learning in the article [10] or evolutionary algorithms in the article [11], have been done. Also, the only research [6] has conducted research on the detection of tooth decay in panoramic photographs. As a result, the question arises that considering the benefits of panoramic photos for patients and professionals and the widespread prevalence of dental caries, is it possible to design a model to detect tooth decay from panoramic photos?

## 2 Materials and Methods

The application of Artificial intelligence (AI) in medicine and dentistry has increased in recent years, which may be seen in the number of published studies [11-19]. In this study, we intend to automate the detection of teeth and tooth decay. In such a way that a neural network is able to detect a complete tooth first in radiographic images of the complete dental view, and then detect whether the tooth is unhealthy or healthy. This system can show the decay rate of the corresponding tooth as a percentage of the

probability of being healthy or the probability of moderate or severe decay next to the enclosing box of each tooth. The general approach of the test includes several steps, we have the following steps in the proposed model:

**First step:** First, the input data is prepared. At this stage, a large number of radiographic images needed for training the network are collected for the network. These images have a specific quality and certain dimensions. In this paper, the dataset used is from [23], which is the Diagnostic Imaging Center of the Southwest State University of Bahia (UFBA-UESC) Dental Images dataset and contains 4000 full dental X-ray images.

**The second stage:** Two labeling process is done to identify the tooth and the other labeling to identify the tooth decay. It should be noted that this stage is a very time-consuming and long process that requires several weeks of effort and cooperation of an expert who is a dentist in this paper. We used the Supervisely tool for tooth detection labeling and for tooth cavity labeling a group of dentists from Mashhad Medical University of Iran tagged the images manually. These data are collected and then divided into two parts, training and testing.

**Third stage:** Training stage: 75% of the data prepared in the previous stage is allocated for training. In this step, identification of image areas and deep learning neural network are used. The proposed regions are identified and introduced by the Region Proposal Network. Then, a decay detector is trained using convolutional neural network. In the following, we will get to know the exact components of this step. At the end of this process, we have a trained neural network model that is ready for the next step, the testing process.

**Fourth stage:** In this stage, 25% of the data that was set aside is now used to test the performance of the network. The result of the training is checked on the specific data of the test. Finally, the network gives us the output of the tooth and whether it is unhealthy, and if it is, how badly it has decay.

Previously, in the reviewed articles, there has been no research on tooth detection, as well as caries detection and, in addition, the severity of caries of each tooth in full-view radiographic images. A limited number of articles have only performed the detection of teeth in the panoramic photo, and only one article [6] has dealt with the detection and identification of tooth decay, which also used genetic algorithms and several other networks and different layers that process is very time-consuming and very complicated.

### 2.1 Data collection and preparation

About 4000 Panoramic dental radiographs were obtained by request from [24]. These photos included all types of decayed, healthy, edentulous teeth, dental veneers, implants, and patients with dental orthodontics. 1000 dental radiograph images were selected from the data, which includes patients with damaged teeth of different severity and edentulous patients with orthodontics. These images were received and tagged with a resolution of 1292\*2440.

### 2.2 Labeling

The labeling operation consists of two steps: First, separating the teeth. Second, labeling the severity of caries.

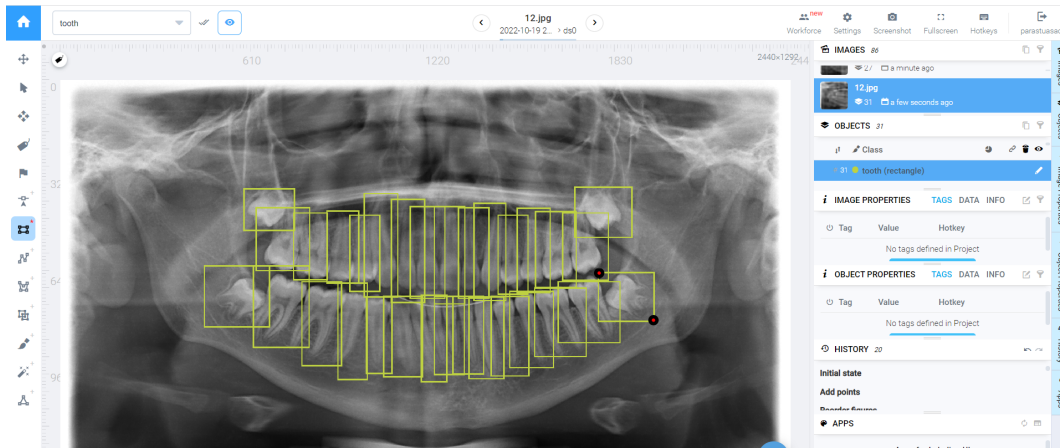


Figure 2: Supervisely tool for labeling

Table 1: ACCURACY AND READABILITY OF THE RESULTS

|              | F0.5 score | Recall | Precision |
|--------------|------------|--------|-----------|
| Trained data | 0.92       | 0.71   | 0.94      |
| Test data    | 0.89       | 0.68   | 0.89      |

Table 2: READING IN THE DIAGNOSIS OF DECAYED TEETH

| dental caries | Recall | dental caries |
|---------------|--------|---------------|
| slight decay  | 0.64   | slight decay  |
| severe decay  | 0.69   | severe decay  |

**2.2.1 Detection of teeth.** There was no need for a dentist specialist to label this part, and we only needed to label the teeth with enclosing boxes, in a way that includes all the components of the tooth from the collar to the root of the tooth.

Various tools have been provided for this purpose. We use the Supervisely tool as can be seen in Fig. 2, which is one of the open source web-based tools, to label the first section. All the images were first uploaded on this site and then the tagging operation was done by a group of people.

**2.2.2 Labeling the severity of caries.** The process of tagging images based on whether they are decayed or not, as well as the degree of decay, is a very long and time-consuming process that requires the efforts of a specialist dentist to make this diagnosis.

The OPG radiograph includes the image of all the teeth in one image, which reduces the accuracy and focus of detecting the caries of each tooth. In the following, we have solved this challenge by presenting a model for labeling dentists. First, there is a need for a quick and easy tool for an expert to use this tool. By using the tkinter and opencv libraries in python, we designed a teeth labeling tool that meets our needs. Fig. 3 shows the window of the designed tool.

Another challenge is the number of 1000 full-view radiographic images, most of which have 32 teeth in two jaws, and the number of

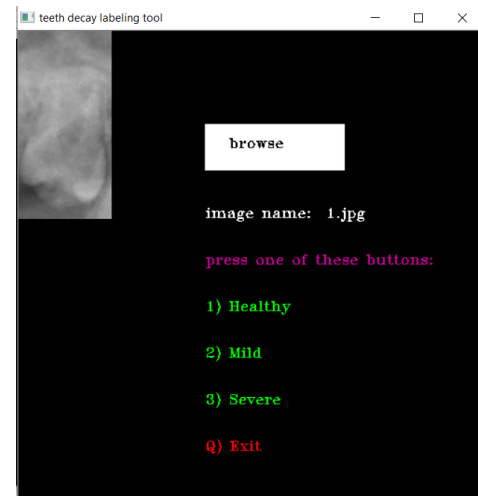


Figure 3: Designed tool window

about 32,000 images of single teeth, which must be carefully labeled by the expert. Here, 750 full view images were used for training data and 250 images were used for testing data. In this section, we asked an expert dental surgeon to label single and enclosed tooth images. He divided the tooth labeling in such a way that each tooth is mapped to one of the healthy, mild caries and severe caries classes.

**2.2.3 Faster Region-Based Convolutional Neural Network (R-CNN) network.** Our proposed approach uses the structure of Faster R-CNN (Fig. 4), which is the most used in R-CNN family networks,

**Table 3: SUMMARY OF PREVIOUS STUDIES [4]**

| Authors                 | Journal                                                            | Country, Year | Variables Detected                  | Image                                 | Total Image Database | Neural Network                                                 | Outcome Metrics                                | Outcome Metrics Value                   |
|-------------------------|--------------------------------------------------------------------|---------------|-------------------------------------|---------------------------------------|----------------------|----------------------------------------------------------------|------------------------------------------------|-----------------------------------------|
| Schwendicke et al. [25] | Journal of Dentistry                                               | England, 2019 | Caries                              | Near-infrared light transillumination | 226                  | Resnet18<br>Resnext50                                          | AUC/Sensitivity/Specificity                    | 0.74/0.59/0.76                          |
| Fukuda et al. [26]      | Oral Radiology                                                     | Japan, 2019   | Vertical root fracture (endodontic) | Panoramic radiography                 | 300                  | DetectNet                                                      | Recall/Precision/F measure                     | 0.75/0.93/0.83.                         |
| Ekert et al. [27]       | Journal of Endodontics                                             | USA, 2019     | Endodontic                          | Panoramic radiography                 | 85                   | CNN                                                            | AUC/Sensitivity                                | Molar: 0.89/0.74 Other teeth: 0.85/0.65 |
| Chen et al. [28]        | Scientific Reports                                                 | England, 2019 | Teeth                               | Periapical images                     | 1250                 | Faster R-CNN                                                   | Recall/Precision                               | 0.728/0.771                             |
| Tuzoff [29]             | Dentomaxillofacial Radiology                                       | England, 2019 | Teeth                               | Panoramic radiography                 | 1352                 | Faster R-CNN                                                   | Sensitivity/Precision                          | 0.9941/0.9945                           |
| Zhang et al. [30]       | Computerized Medical Imaging and Graphics                          | USA, 2018     | Teeth                               | Periapical images                     | 700                  | Faster-RCNN/region-based fully convolutional networks (R-FCN). | Precision/Recall                               | 0.958/0.961                             |
| Raith et al. [31]       | Computers in Biology and Medicine                                  | England, 2017 | Teeth                               | -                                     | -                    | ANN                                                            | Performance                                    | 0.93                                    |
| Srivastava et al. [32]  | NIPS 2017 workshop on Machine Learning for Health (NIPS 2017 ML4H) | USA, 2017     | Caries                              | Bitewing                              | 3000                 | FCNN (deep fully convolutional neural network)                 | Recall/Precision/F1-Score                      | 0.805/0.615/0.7                         |
| Jader et al. [33]       | IEEE                                                               | Brazil, 2018  | Teeth                               | Panoramic radiography                 | 1500                 | Mask R-CNN                                                     | Accuracy/F1-score/Precision/Recall/Specificity | 0.98/0.88/0.94/0.84/0.99                |
| Miki et al. [34]        | Computers in Biology and Medicine                                  | USA, 2017     | Teeth                               | Cone-beam computed tomography (CT)    | 52                   | AlexNet                                                        | Accuracy                                       | 0.88                                    |
| Veleminská [35]         | Anthropologischer Anzeiger                                         | Germany, 2013 | Teeth                               | Panoramic radiography                 | 1393                 | RBFNN<br>GAME                                                  | Accuracy                                       | -                                       |

|                             |                                                                        |                            |        |                                 |      |                                                                 |                           |                                                                                       |
|-----------------------------|------------------------------------------------------------------------|----------------------------|--------|---------------------------------|------|-----------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------|
| Casalengo et al. [36]       | Journal of Dental Research                                             | USA, 2019                  | Caries | Near-infrared transillumination | 217  | CNNs for semantic segmentation                                  | AUC                       | 0.836 and 0.856 for occlusal and proximal lesions, respectively                       |
| Lee et al. [37]             | Journal of Dentistry                                                   | England, 2018              | Caries | Periapical                      | 3000 | Deep CNN algorithm, weight factors                              | Accuracy/AUC              | premolar, molar, and both premolar and molar: 0.89, 0.88, and 0.82/0.917, 0.89, 0.845 |
| Zanella-Calzada et al. [38] | Bioengineering                                                         | Switzerland, 2018          | Caries | -                               | 9812 | ANN                                                             | Accuracy/AUC              | 0.69/0.75                                                                             |
| Muramatsu et al. [39]       | Oral Radiology                                                         | Japan, 2020                | Teeth  | Panoramic radiographs           | 100  | Object detection network using fourfold cross-validation method | Sensitivity/Accuracy      | 0.964/0.932                                                                           |
| Prajapati et al. [40]       | 5th International Symposium on Computational and Business Intelligence | United Arab Emirates, 2017 | Caries | Radiovisiography images         | 251  | CNN                                                             | Accuracy                  | 0.875                                                                                 |
| Oktay, A. [41]              | IEEE                                                                   | Turkey, 2017               | Teeth  | Panoramic radiographs           | 100  | AlexNet                                                         | Accuracy                  | >0.92                                                                                 |
| Geetha et al. [42]          | Health Information Science and Systems                                 | Switzerland, 2020          | Caries | Intraoral radiographs           | 105  | Back-propagation neural network                                 | Accuracy/Precision recall | 0.971/0.987                                                                           |

as the basic neural network to estimate the position of teeth and also to detect tooth decay. Faster R-CNN network was introduced for the first time in 2015. In the rest of the articles of the R-CNN family, the evolution has always been related to the computational efficiency and reducing the execution time, as well as increasing the accuracy. These networks usually include 4 sections.

- 1-Algorithm suggesting Region Proposal Network (RPN) region to introduce bounding boxes that may contain objects.
- 2- Feature generation to extract features from these objects
- 3- A separator layer that has the task of mapping each object to its corresponding class.
- 4- A regression layer that improves the location of the detected bounding boxes.

#### *Extraction of suggested areas with RPN network*

Faster R-CNN uses the RPN network to suggest areas. In this network, the images first pass through a trained convolution network, the output of the pre-trained network is a map of the features of the input images.

The output of the pre-trained network is entered into a region proposal network, where the input of this network is the feature image and the output of those proposed regions. The proposed areas of rectangles are finally given to a ROI pooling layer network. This network is the same pyramid pooling network with pyramid level.

#### *Fully Connected Layers*

The output of the Region of Interest (ROI) pooling network with the dimensions (number of proposed areas\*length\*width\*number of features) is entered into a network including 2 fully connected layers. It is placed for the purpose of regression. The output of the

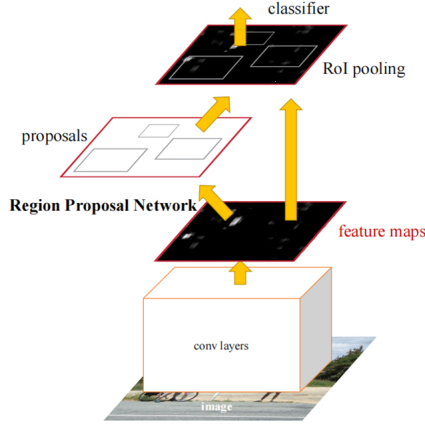


Figure 4: Faster r-cnn network structure [32]

separator network shows the probability of each class of objects and the output of the Regressor network shows the final location of the enclosing boxes, which is the improved output of the RPN network. In order to separate the outputs of the fully connected layer related to separation, they are entered into a Softmax layer under the title of input.

### 3 Experiments and results

#### 3.1 detector system training process

The learning steps are as follows:

First, the RPN network was trained separately. The weights of the convolutional network are preset on ImageNet and are only fine-tuned for the purpose of region proposal.

Second, FAST R-CNN diagnostic network was trained separately. The weights of the CNN network are preset and fine-tuned only for the purpose of region proposal.

Third, The RPN network was initialized with the values learned in step 1, and the weights of the CNN network are already set and fixed, and only the weights of the RPN network are fine tuned.

Fourth, this time, by using the learned RPN, the fast R-CNN detector is fine-tuned, that is, all other network weights remain constant. The result of this test is a faster R-CNN detector that shared the weights of the convolutional network.

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (1)$$

The arrangement of convolution and maxpooling layers continuously follows the entire architecture in VGG16 (Fig 5). At the end, it has two fully connected layers, followed by a softmax for the output. The RPN network generates the desired region for the input radiographic images. RPN is a fully convolutional network that includes 3 convolutional layers and one recommender layer. Tooth emergence is checked by regression and separation by bounding boxes by RPN network.

#### 3.2 loss function

The loss function is defined as follows:

$$t_x^* = \frac{x^* - x_a}{w_a} \cdot t_y^* = \frac{y^* - y_a}{h_a} \cdot t_w^* = \log\left(\frac{w^*}{w_a}\right) \cdot t_h^* = \log(h^*/h_a) \quad (2)$$

which in this formula  $i$  represents the index of small groups,  $P_i$  represents the probability estimate of the box with constant ratios and  $P_i^*$  the label of the reference data, is a representation of the vector,  $t_i^*$  the accuracy box of the reference data that is related to the box with positive constant ratios,  $L_{cls}$  the function Separator loss is on two classes (being an object and not being an object) and  $L_{reg}$  is a regressor loss function.

$L_{reg}$  is activated only when the input box must contain the object, which in this case  $P_i^*$  is data reference 1.  $t_i$  is the predicted output of the regressor layer and contains four variables.  $t_x, t_y, t_w, t_h$  The regression target  $t_i^*$  is calculated as follows:

$$t_x^* = \frac{x^* - x_a}{w_a} \cdot t_y^* = \frac{y^* - y_a}{h_a} \cdot t_w^* = \log\left(\frac{w^*}{w_a}\right) \cdot t_h^* = \log(h^*/h_a) \quad (3)$$

which in these equations  $h, w, y, x$  show the coordinates of the center of the box as well as its length and width. The variables  $X^*, X_a$  are respectively for the  $h, w, y$  predicted box and the box with fixed dimensions, as well as the reference data box.

#### 3.3 Results of the proposed method on the test data set:

As mentioned above, 25% of the input data was discarded as test data. In order to evaluate the accuracy of the proposed model and finally compare it with the previously presented approaches, we use three accuracy measurement criteria: The accuracy criterion of Table 1 is the ratio of correct positives to the total number of detected positives. It is used in order to check the accuracy in detecting correct positive and false positive sections. The readability criterion of Table 1 is the ratio of correct positives to the sum of correct positives and false negatives. In our problem, the high readability value indicates that the proposed approach in false negatives (teeth that were healthy and incorrectly labeled as broken) number It has little to do with correct diagnoses. Since falsely diagnosed healthy decayed teeth are more important than false positives (teeth that were healthy and falsely labeled as decayed), we use the f 0.5 criterion should focus on minimizing the number of false positive diagnoses). Therefore, in order to increase the weight on accuracy and reduce the importance of readability, we used the f 0.5 score as the best criterion to measure the efficiency of the model [6]. Table 2 and Table 3 shows the results that has been achieved.

### 4 Conclusion and future works

This study validates the hypothesis that a Faster RCNN-based system can be trained to detect teeth and caries in full-view radiographs for automated dental mapping purposes. System performance is close to expert level, meaning the radiologist can use the system output for automated charting when only minor evaluations and corrections are required instead of manual data entry. Since this is a proof-of-concept study, there is potential to increase system performance. The results of tooth and caries detection can be improved or strengthened by implementing additional techniques. Recent

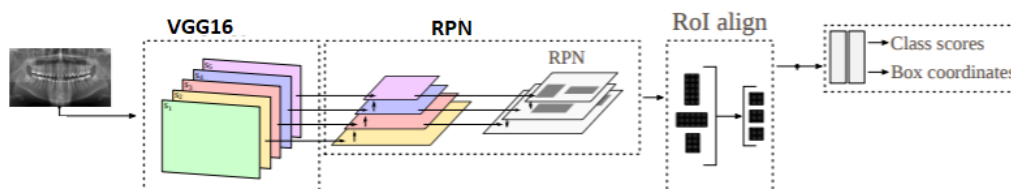


Figure 5: Proposed model of tooth and tooth decay detector

CNN architectures for extracting and classifying features based on the obtained results, it can be concluded that artificial intelligence deep learning algorithms have the potential to further explore their applications and implement them in clinical dental environments. These full-view images are of lower quality and resolution than periapical and bitewing images, which makes the process of detecting tooth decay, especially for anterior teeth, challenging even for dentists. Several articles have even compared the efficiency of these automatic tooth and caries detection networks with the performance of dentists, both beginners and experts. This issue is not discussed in this paper, but there is room for more research and investigation in this area.

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